

Samuel Silverman

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Education

PhD, Computer Science 2021-Present
Boston University, Boston, MA
Advisor: Emily Whiting

MS, Computer Science 2019-2021
Columbia University, New York, NY

BS, Computer Science 2015-2019
University of Massachusetts Amherst, Amherst, MA

Publications

Dylan J. Balter, Colin McMillen, Alec Ewe, Jonathan Thomas, Samuel Silverman, Latitha Parameswaran, Luis Fernando Velásquez-García, Emily Whiting, Steven Patterson, Hilmar Kerner, Keith A. Brown. 2026. *Giant Flexoelectricity of Additively Manufactured Polylactic Acid*. **Additive Manufacturing**.
<https://doi.org/10.1016/j.addma.2025.105066>.

Samuel Silverman, Dylan Balter, Keith A. Brown, Emily Whiting. 2025. *Random-Walk Microstructures for Differentiable Topology Optimization*. **Proceedings of the 10th Annual ACM Symposium on Computational Fabrication**.
<https://doi.org/10.1145/3745778.3766645>.

Samuel Silverman, Kelsey L. Snapp, Keith A. Brown, Emily Whiting. 2025. *Data-Driven Nonlinear Deformation Design of 3D-Printable Shells*. **3D Printing and Additive Manufacturing**.
<https://doi.org/10.1089/3dp.2024.0175>.

Kelsey L. Snapp, Samuel Silverman, Richard Pang, Thomas M. Tiano, Timothy J. Lawton, Emily Whiting, Keith A. Brown. 2024. *A Physics-Informed Impact Model Refined by Multi-Fidelity Transfer Learning*. **Extreme Mechanics Letters**.
<https://doi.org/10.1016/j.eml.2024.102223>.

Kelsey L. Snapp, Benjamin Verdier, Aldair E. Gongora, Samuel Silverman, Adedire D. Adesiji, Elise F. Morgan, Timothy J. Lawton, Emily Whiting, Keith A. Brown. 2024. *Superlative Mechanical Energy Absorbing Efficiency Discovered Through Self-Driving Lab-Human Partnership*. **Nature Communications**.
<https://doi.org/10.1038/s41467-024-48534-4>.

Vishnu Nair, Jay L. Karp, Samuel Silverman, Mohar Kalra, Hollis Lehv, Faizan Jamil, Brian A. Smith. 2021. *Navstick: Making Video Games Blind-Accessible via the Ability to Look Around*. **Proceedings of the 34th Annual ACM Symposium on User Interface Software and Technology**.
<https://doi.org/10.1145/3472749.3474768>.

Employment

Charles River Analytics

2021

Software Engineer Intern

Member of the Sensing, Perception, and Applied Robotics Division. Implemented algorithms to estimate detected objects' real-world positions and dimensions using data from a smart maritime camera. Built a pipeline to convert trained PyTorch, TensorFlow, XGBoost, and LightGBM classifiers from Python to C++ for efficient deployment.

Open-Source Software

Monad

C++ library for homogenization of structured 2D and 3D grids. It provides solvers and material models for computing effective (homogenized) material tensors from heterogeneous microstructures.

rwmicro

C++ library for generating random-walk microstructures—binary patterns created by a periodic random walk.

gcs

Python library for generating 3D meshes of generalized cylindrical shells.

Invited Talks

Data-Driven Nonlinear Deformation Design of 3D-Printable Shells

October 24, 2025

Materials Day 2025

Boston University, Boston, MA

Data-Driven Nonlinear Deformation Design of 3D-Printable Shells

March 8, 2023

Computational Fabrication Group

University of Washington, Seattle, WA

Data-Driven Nonlinear Deformation Design of 3D-Printable Shells

March 27, 2023

Transformative Robotics Lab

Northeastern University, Boston, MA

Teaching

Teaching Assistant

Spring 2026

CAS CS 581: Computational Fabrication

Boston University, Boston, MA

Teaching Assistant

Fall 2025

CAS CS 320: Concepts of Programming Languages

Boston University, Boston, MA

Teaching Assistant

Spring 2024

CAS CS 581: Computational Fabrication

Boston University, Boston, MA

Teaching Assistant

Summer 1 2023

CAS CS 411: Software Engineering

Boston University, Boston, MA

Course Assistant

Fall 2020

COMS 4170: User Interface Design

Columbia University, New York, NY

Awards

Dean's Fellowship, Boston University

2021-2022

Press Coverage

The Brink. *3D Printing Robot Uses AI Machine Learning for US Army Research*

May 2024